7. Fixed-Size Collection -- Arrays

Example 7.1

```
// Example Java code on the book
public class Example7_1 {
    public static void main(String[] args) {
        int[] scores = {80, 90, 75, 60};

        int sum = 0;
        for (int i = 0; i < scores.length; i++) {
            sum += scores[i];
        }

        int average = sum / scores.length;

        String result = (average >= 70) ? "Pass" : "Fail";

        System.out.println("Average: " + average);
        System.out.println("Result: " + result);
    }
}
```

Covered concepts: array, **for loop**, conditional operator

Example 7.2

```
// Example Java code on the book
public class Example7_2 {
    public static void main(String[] args) {
        int[] daysInMonth = {31,28,31,30,31,30,31,31,30,31,30,31};

        int month = 2; // February
        System.out.println(daysInMonth[month - 1]);
    }
}
```

Covered concepts: array, lookup table

7.1. Array

Definition

An array is a fixed-size collection of elements of the same type.

Features

- Elements have the same data type

- Elements are stored in order
- Index starts from `0`
- Size cannot change after creation

Example

```
int[] scores = {80, 90, 75};
```

- `scores` stores three integers
- `scores[0]` is `80`

Remark

- Index out of range causes runtime error

7.2. Array Index

Definition

An index is the position of an element in an array.

Features

- First index is `0`
- Last index is `length - 1`

Example

```
scores[1] = 95;
```

- Changes the second element

Remark

- Invalid index causes `ArrayIndexOutOfBoundsException`

7.3. for Loop（重点）

Definition

A for loop repeats code a fixed number of times.

Features

- Has initialization, condition, update
- Very common with arrays

7.3.1. for Loop Structure

Standard loop form

```
for (int i = 0; i < n; i++) {
    // loop body
}
```

- `i` is the loop variable

7.3.2. for Loop with Array

Used to visit every element

```
for (int i = 0; i < scores.length; i++) {
    System.out.println(scores[i]);
}
```

- `length` gives array size

Remark

- Most array problems use a for loop

7.4. Conditional Operator

Definition

The conditional operator chooses one of two values.

Features

- Also called ternary operator
- Short form of if-else

Example

```
int max = (a > b) ? a : b;
```

- If `a > b`, result is `a`

Remark

- Use only for simple conditions

7.5. Lookup Table

Definition

A lookup table uses an array to store predefined values.

Features

- Fast access by index
- Avoids long if-else chains

Example

```
int[] days = {31,28,31,30};
System.out.println(days[1]);
```

- Index gives direct result

Remark

- Very common in exams and practice

7.6. Array Length

Definition

`length` gives the number of elements in an array.

Features

- Used in loops
- Prevents index errors

Example

```
for (int i = 0; i < arr.length; i++) {
    // safe access
}
```

Remark

- `length` is a field, not a method